Rheeya Uppaal

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EDUCATION

University of Wisconsin-Madison	Doctor of Philosophy in Computer Science	2022-Present
University of Massachusetts, Amherst	Master of Science in Computer Science	2017-19
Symbiosis Institute of Technology	Bachelor of Technology in Computer Science	2013-17

RESEARCH EXPERIENCE

AWS (Amazon) AI Labs New York, June-August 2025
Applied Scientist Intern, under Dr Phu Mon Htut

Amazon Science Seattle, June-August 2024
Applied Scientist Intern, under Dr Markus Dreyer and Prof Mohit Bansal

Goldman Sachs CoreAI New York City, 2019 – 2021
Research Engineer, *under Dr Vijay Saraswat*

UMass Amherst Amherst, 2018-2020
Research Assistant, *under Prof Andrew McCallum and Prof Madalina Fiterau*

Microsoft Research Montreal Remote, February-July 2018
Research mentee, under *Dr Timothy Hazen*

SELECTED PUBLICATIONS

- *Journey Before Destination: On the importance of Visual Faithfulness in Slow Thinking.* Under review at ACL Rolling Review.
- *Model Editing as a Robust and Denoised variant of DPO: A Case Study on Toxicity.* In the Thirteenth International Conference on Learning Representations (**ICLR**), 2025. [Talk][Article]
- *How Useful is Continued Pre-Training for Generative Unsupervised Domain Adaptation?* In the 9th Workshop on Representation Learning for NLP (Rep4NLP) at the 62nd Annual Meeting of the Association for Computational Linguistics (**ACL**), 2024.
- *Is Fine-tuning Needed? Pre-trained Language Models Are Near Perfect for Out-of-Domain Detection.* In the 61st Annual Meeting of the Association for Computational Linguistics (**ACL**), 2023. [Talk] [Media coverage]
- *Long Document Summarization in a Low Resource Setting using Pretrained Language Models*. In the Student Workshop at the Joint Conference of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (**ACL-IJCNLP**), 2021.
- *Overcoming Practical Issues of Deep Active Learning and its Applications on Named Entity Recognition.* In European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (**ECML-PKDD**), 2020 (Journal Track).

INVITED TALKS

- From Internals to Outcomes: Enhancing Truthfulness and Safety in Language Models. At Freestone Grove, 2025.
- Towards Robust Model Editing Techniques. Cohere for AI's Research Connections Community, 2025.
- Model Editing for Aligning Models to Preferred Behaviours. Amazon Science GPS Reading Group, 2024
- Multi-task Learning in NLP. UW Madison's CS 769: Advanced Natural Language Processing, Fall 2024.
- Reducing Toxicity in Language Models through Weight Editing. ML-X Forum at UW Madison, 2024.

SELECTED AWARDS, SERVICE AND OUTREACH

- **Winner**, UW Madison First-year CS Departmental Scholarship, 2022.
- **Winner**, *Karamveer Chakra Award* (instituted by iCONGO and the **United Nations**), 2021.
- **Organizing:** *Carpentries* workshop on *Trustworthy AI: Validity, Fairness, Explainability, and Uncertainty Assessments*, Fall 2024.
- **Mentoring:** *ML-X Machine Learning Marathon* (2023), *UW Madison ACM-W Year Long Mentorship Program* (2022-), *UMass Amherst MS Clusters* (2021), *UMass Industrial Independent Study* 2020 (representing Goldman Sachs).
- **Service:** Leadership of the research team of the Wisconsin AI Safety Initiative. Worked with the NGOs 'Swacch', 'Goonj' and 'ALL' to tackle multiple social and environmental issues, including organizing charity donations and welfare activities.

TECHNICAL SKILLS: Languages: Python | Deep Learning Frameworks: PyTorch, HuggingFace